

Kian – Machine Learning Engineer | Real-Time Decision Systems

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PROFILE

Machine learning engineer specialising in real-time decision systems, probabilistic modelling, latent state estimation, anomaly detection and reliability monitoring. Experience designing and deploying live inference pipelines for noisy, non-linear data under real-world operational constraints. Focusing on operational robustness and practical implementations. Core background in mathematics and data science and regulated operational settings, combining quantitative reasoning with engineering ownership and commercial awareness.

HIGHLIGHTED PROJECTS (See GitHub for full project overview)

Real-time Market Intelligence Cloud Console (2021, 2024 - 2025)

- Designed and deployed a continuously running analytics platform integrating on-chain, EVM and off-chain market data, for live monitoring and signal generation.
- Built a probabilistic time-series pipeline for dynamic market conditions, focusing on noisy inputs and handling uncertainty for system reliability.
- Deployed onto GCP for autonomous, low latency ingestion and continuous monitoring across multiple live data sources and remote access.
- Fully integrated a Python back-end to an API-driven workflow, supporting structured downstream analysis and decision making under uncertainty.

Probabilistic Latent Temperature Field Inference (2025)

- Developed a probabilistic reconstruction workflow, for latent temperature fields from sparse, noisy and incomplete sensor observations for simulated server-room conditions.
- Applied PDE-based inference to estimate hidden spatial structure under incomplete measurements. Improving robustness and generalisation rather than pure data-driven interpolation.
- Evaluated reconstruction performance under sparse and noisy sensor conditions to assess behaviour as monitoring reliability degraded.

Quantitative Comparison of Manifold-based and Remote Sensing clusters (2026)

- Designed and built a geospatial quantitative workflow, using Python, SQL and Google Earth Engine to incorporate manufacturing, remote-sensing and open street data over a common raster in China.
- Compared manifold-learning and remote-sensing clustering outcomes using PCA, UMAP, HDBSCAN, ARI and Moran's I to quantify concordance, stability and spatial consistency.
- Developed a dual-linked spatial visualisation and filtering workflow to support diagnostic inspection of cluster assignment and grid-cell attributes across manufacturing regions.

PROFESSIONAL EXPERIENCE

Property Development & Operations (2024 – 2025)

- Led end-to-end acquisition and operational optimisation of a residential asset under strict capital and regulatory constraints, improving valuation and increasing gross annual yield while balancing financial risk.
- Managed budgeting, compliance requirements, contractor coordination and decision making across the project lifecycle under real-world time constraints.
- Strengthened commercial awareness, prioritisation and operational ownership in a demanding environment.

Personal Website Development (2023-2024)

- Designed and deployed web systems for local businesses, integrating API-driven workflows and performance monitoring, to improve operational reliability and recurring revenue stability.
- Delivered practical solutions focusing on usability, maintainability and long term business value rather than a one off prototype.
- Independent project management from requirements gathered through implementation and deployment.

Lloyds Banking Group - Customer Service Representative (2022)

- Worked within a regulated, high-volume financial environment, handling transactions and compliance-bound account operations.
- Developed understanding of operational risk, transaction monitoring and structured financial processes.
- Built experience operating accurately under service constraints in a customer-facing setting.

TECHNICAL SKILLS**Languages & Tools**

Python (FastAPI, NumPy, Pandas, GeoPandas, OpenCV, scikit-learn, TensorFlow)

Java, JavaScript, SQL, Solidity (EVM), Cloud deployment on GCP, API integration

Core Concepts

Probabilistic modelling, State-space and time series modelling, Anomaly detection, Dimensionality reduction, Bayesian Inference, and Reliability analysis, System monitoring.

Other Skills

Project Ownership, Investor Communication, Business Operations and Decision-making